Not In Our Time: Transport-Based Content Selection

An Inspectable, Manipulable Alternative to RAG — Demonstrated on Literary Discussion Podcasts for 15 Victorian Novels

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"Fog everywhere. Fog up the river, where it flows among green aits and meadows; fog down the river, where it rolls defiled among the tiers of shipping and the waterside pollutions of a great (and dirty) city. Fog on the Essex marshes, fog on the Kentish heights." — Charles Dickens, *Bleak House* (1853), Chapter 1

We prompt an LLM to build **literary discussion podcasts** in the style of BBC Radio 4's *In Our Time*: a host and three expert panellists discuss a novel, grounding their conversation in specific passages from the text. The host prepares by interviewing each expert privately, then steers the live discussion with targeted questions — producing cross-examination, not parallel monologue. This a simple **neurosymbolic** system: the process of podcast planning is explicitly engineered, the outcome is co-determined by LLM choices. Everything from passage selection to script generation is inspectable and adjustable. This poster asks what editorial choices matter, and whether their effects are predictable.

The Questions

RAG selects passages by embedding similarity — effective, but opaque. You cannot see *why* a passage was chosen, adjust selection criteria, or explore how different editorial priorities reshape the output.

We run **min-cost flow (transport)** and embedding retrieval side-by-side on 15 Victorian novels, with a no-passages baseline, to answer four questions:

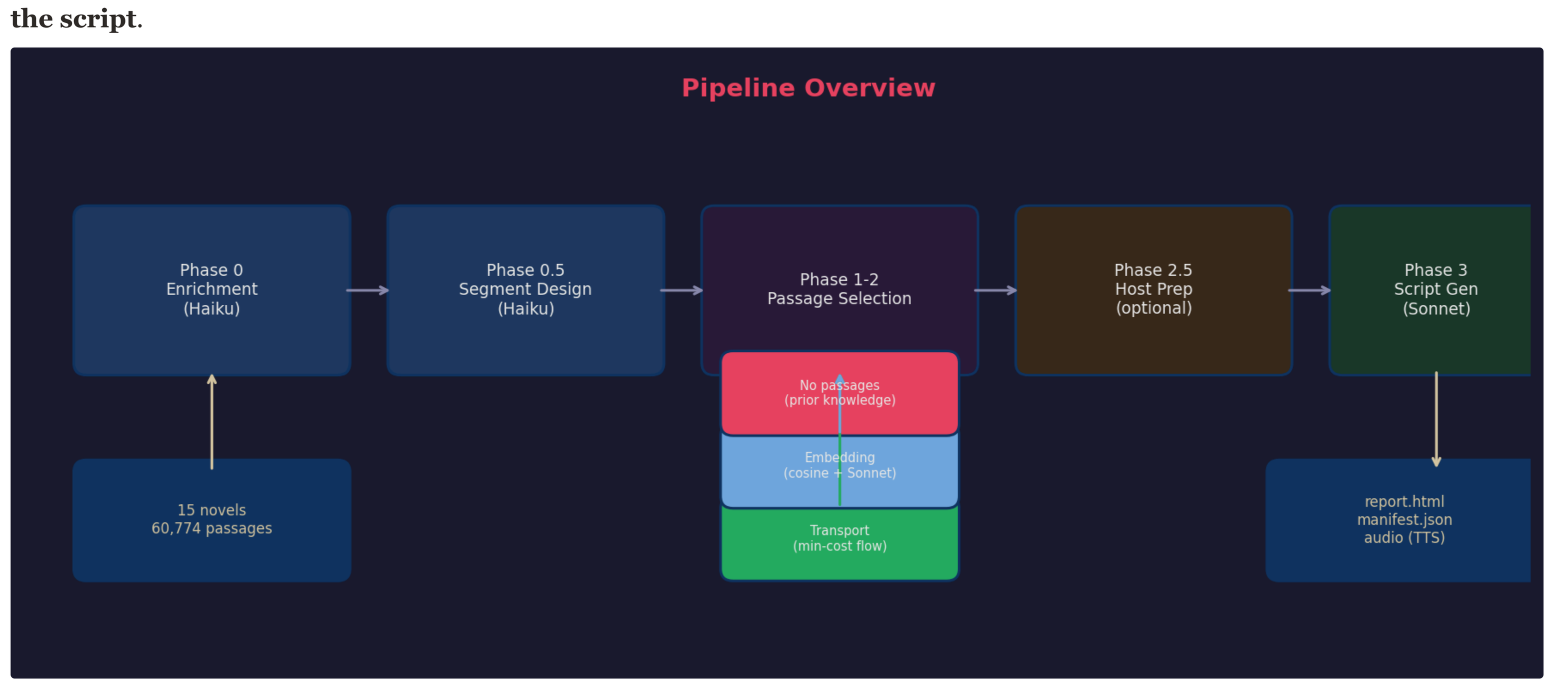
1. **Does selection strategy affect discussion quality?**
2. **Does passage grounding prevent confabulation?**
3. **What does transport offer that embedding cannot?**
4. **What conversational scaffolding produces genuine discussion?**
5. **Are the relative effects of editorial choices predictable?**

15	60,774	180
novels	passages	conditions

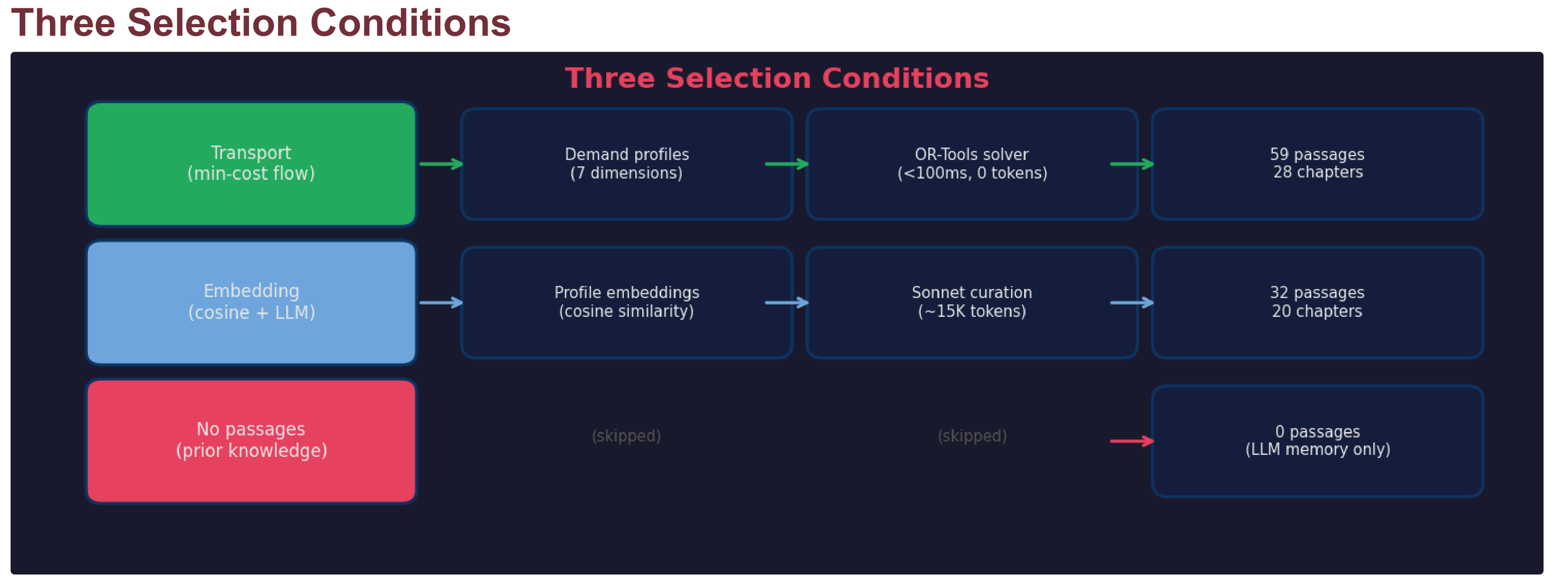
Seven authors (Dickens, Eliot, Gaskell, Forster, Collins, Gissing, Oliphant), seven decades (1850–1924), from the very well-known to the obscure.

The Pipeline

Each episode passes through five stages: the novel is **enriched** passage by passage; an LLM **designs segments**; passages are **selected** by one of three strategies; experts are **prepared** through pre-interviews and question planning; a final LLM call **generates the script**.



Three Selection Conditions



Host Preparation

Before each segment is generated, every expert is pre-interviewed privately: *what strikes you? where would you disagree?* The host reads all responses and plans 3–5 targeted questions per segment. Question density rises from ~1 to ~11 per segment across all 15 novels — producing cross-examination, not parallel monologue.

Speech Generation

Scripts are rendered to audio using Gemini TTS with per-speaker voice assignment. Delivery annotations are embedded in the script itself — accent direction, pacing, pause markers, and emphasis are part of the structured output, specified utterance by utterance, not added in post-production.

Each expert gets a consistent voice across all segments of an episode. Switching the expert panel automatically switches the voices: the voices are properties of the personas, not of the episode.

Provenance Badges

Every page on this site carries a provenance badge — a live classification of how much of its content was written by AI. Click the badge in the nav bar to see the commit history.

The mechanism reads git history. When Claude helps write a commit, the log records a `Co-Authored-By: Claude` tag. Then `git blame` measures what fraction of current lines trace back to AI commits. Commits with human-edit signals in the message count toward human authorship — creating a direct incentive: revise substantively, and your badge improves.

Three verdicts: **AI-drafted** — most lines from AI commits with no human-edit signals; **AI-drafted, human-edited** — substantial revision on top; **Effectively human-written** — majority of lines from human commits.

This poster: 9 AI commits of 9 total. Verdict: **AI-drafted**.

Does Selection Method Matter?

Transport and embedding select almost entirely different passages — fewer than 6% in common (Jaccard J = 0.057, Bleak House). Yet the discussions they produce are nearly identical by every structural measure.

Transport	J = 0.057	Embedding
59 passages, 28 chapters	<6% overlap in passages	32 passages, 20 chapters
Character & plot focused		Critique & theme focused
Min-cost flow solver		Cosine similarity + LLM

Yet: Expert Airtime Stable to ±2%

Eleanor Hartley (literary critic): **32–37%**   James Blackstone (social historian): **30–37%**   Caroline Woodcourt (close reader): **29–34%**

The pipeline reinforces expert identity at every stage — enrichment dimensions match demand profiles, the selector satisfies those demands, host preparation targets each expert, and the script prompt embeds their perspective. **Framework dominates strategy.** This is suggestive of Fish's (1980) interpretive communities thesis — but the engineering is the mechanism, not the readers.

Host Preparation

If selection method barely matters, what does? Before each segment is generated, every expert is pre-interviewed privately: *what strikes you about these passages? where would you disagree with the other panellists?* The host reads all responses and plans 3–5 targeted questions designed to surface disagreement and draw out cross-references between experts.

~1	~11	5–14×
questions/segment without prep	questions/segment with prep	range across 15 novels

Without preparation, the host asks generic questions and experts deliver monologues. With it, the host asks sharp, targeted questions and the experts talk *to each other*. The effect is uniform across all 15 novels — it is a property of the scaffolding, not of the material.

Like transport's cost-and-constraint structure, host preparation is fully inspectable: the pre-interview responses, the planned questions, and the resulting turns are all readable in the episode report. Adjust the pre-interview prompts, and a different conversation follows.

Two Panels, Same Novel

Literary Panel

Eleanor Hartley (Literary Critic):
"What strikes me about that Nemo sentence is the grammatical audacity of it. 'A deserted infant' — Dickens reaches for the most helpless possible image of erasure. The infant image should be about beginning, and Dickens flips it into a closing."

James Blackstone (Social Historian):
"The fog is not decoration. It is the Court of Chancery made visible. 'Lies there with no more track behind him that any one can trace than a deserted infant.' That sentence is the whole novel compressed into a single image."

Caroline Woodcourt (Close Reader):
"Readers sometimes use the fog as a kind of permission — 'this is a great symbolic novel, I'm in the presence of Literature with a capital L' — and then they stop actually listening to what Dickens is telling them. What moves me is something far quieter — the moment when a human life is described as leaving no trace."

Interdisciplinary Panel

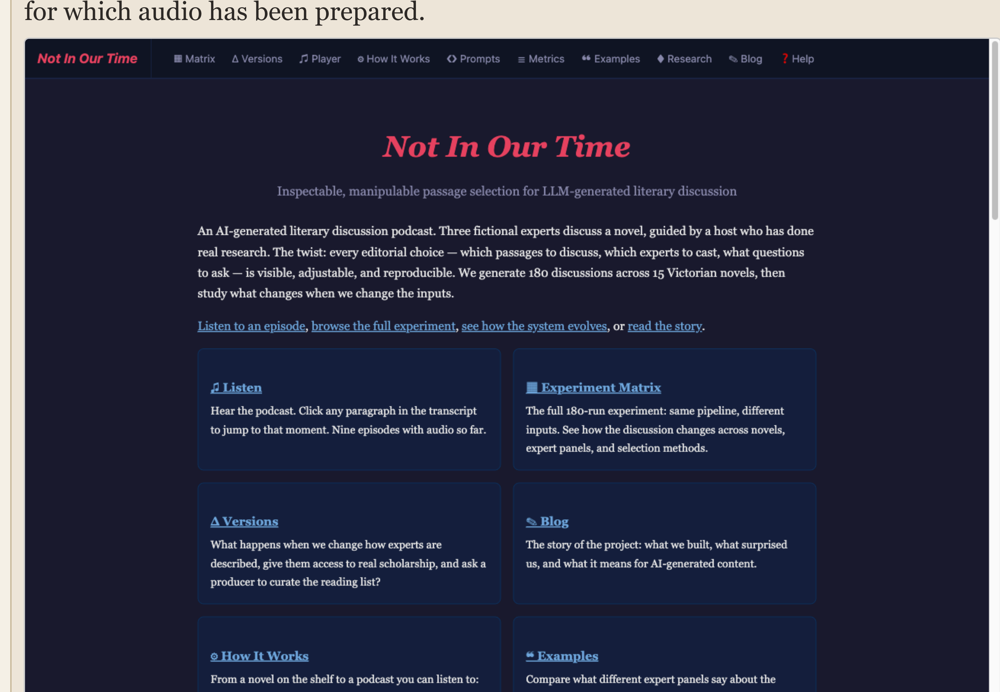
Sarah Chen (NLP Scientist):
"That is a noun phrase with four pre-nominal modifiers, each compounded — and then 'brazen-faced' pulls the floor out. The first three feel like nautical specifications — they're doing the register of technical competence — and the fourth is an insult."

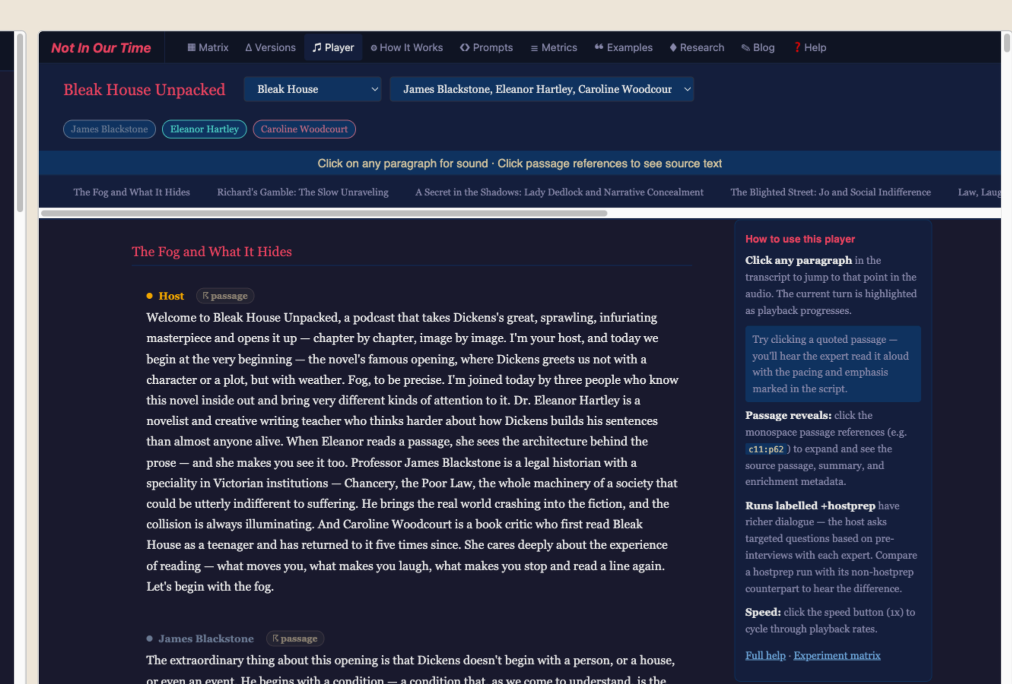
Rebecca Martinez (Astronomer):
"The ships of Law and Equity aren't sunk, they're in reserve. In my field, we'd say the system is in a quasi-stable state. The energy is there. Nothing has been resolved. The fog is the natural atmosphere of a system that has never actually resolved anything."

The experts bring their **metaphors** (quasi-stable states, leitmotifs, phrase structure) but not their **methods** — the scaffold imports literary-critical norms regardless of the persona.

Interactive Demo

Each *condition* is a specific combination of novel, expert panel, grounding strategy, and host-preparation setting — 180 in total. Browse any condition and listen to those for which audio has been prepared.





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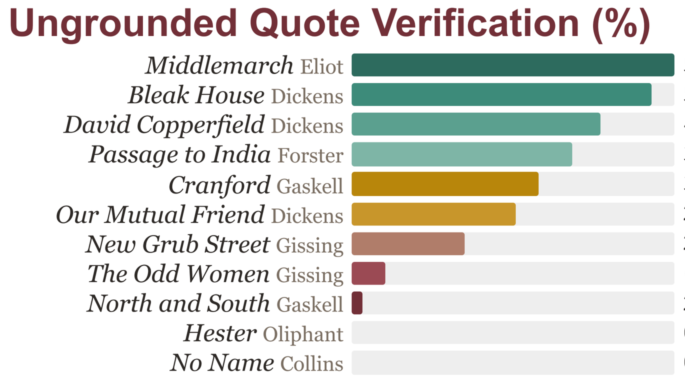
The Knowledge Gradient

Without passage grounding, the LLM produces graduated pastiche — not hallucination. Fidelity tracks novel prominence.

82% of invented quotes draw 90–100% of their words from the novel's own vocabulary. Mean overlap: 95.1%.

Ungrounded Quote Verification (%)



11 of 15 novels shown; omitted novels (Hard Times 39%, Mill on the Floss 37%, Daniel Deronda 30%, Miss Marjoribanks 3%) follow the same pattern. Grounded conditions achieve **97–98%** for *all* novels. The gradient reflects analytical reinforcement in training data (Carlini et al., 2021, 2023), not prose style.

Grounding vs No Passages

Quote verification measures whether a quoted passage can be found verbatim in the novel. **Transport** (min-cost flow, grounded): **98.0%**. **Embedding** (retrieval + LLM, grounded): **97.2%**. **No passages** (prior knowledge only): **48.9%**.

The two grounded conditions converge on near-perfect fidelity regardless of which passages were selected. No-passages conditions fall sharply — not to zero, but to the level of the knowledge gradient: novels the LLM knows well score higher, novels it barely knows score near zero.

Pastiche in Action

Pastiche is imitation of style, not reproduction of text. Ungrounded outputs draw on training-data memory: the vocabulary and register are absorbed from the novel; the sentences were never written by the author.

From a no-passages run on *Hester* (Oliphant, 1883) — a novel with 0% verified grounding. Every quoted sentence was invented by the LLM.

> *"The two men who had loved her — or rather who had shown her some passing tenderness — had both given her a great deal to think of, but neither of them satisfied her mind."* **Invented**
> Oliphant's qualifying irony absorbed from training data; no such sentence exists in the novel.

Contrast: grounded run on *Bleak House* (Dickens, 1853) — 98% verified.

> *"It is a melancholy truth that even great men have their poor relations."* **Verified**
> Traced verbatim to Chapter 28 via passage grounding.

Answers

1. No — strategy barely matters for quality.
Transport and embedding select <6% overlapping passages (J = 0.057) yet produce discussions with identical expert airtime (±2%). Expert personas dominate so strongly that the raw material is nearly irrelevant — *framework dominates strategy.*

2. Yes — grounding prevents confabulation.
Grounded conditions achieve 97–98% verified quotes for *all* novels. No-passages conditions fall to 49% overall — and to 0% for novels the model barely knows. The risk is pastiche, not hallucination, but it is unreliable.

3. Manipulability — what embedding cannot offer.
Every editorial choice is a cost, a flow, a constraint. Solving takes <1 s and zero LLM tokens. Adjust one weight, add one constraint, swap a persona — and inspect what changes. Transport turns a black-box pipeline into a navigable configuration space.

4. Scaffolding is decisive, not emergent.
Host preparation raises questions from ~1 to ~11 per segment, uniformly across all 15 novels. Conversational dynamics are designed — a property of the prompt architecture, not the LLM.

5. Manipulable, up to a point
This is a neurosymbolic system. We can vary the symbolic part and explore, but the outcome is still co-determined by opaque LLM tendencies.

Framework > Grounding > Strategy
A clear hierarchy. Framework determines character. Grounding gates accuracy. Strategy is interchangeable on quality — but only transport makes that interchangeability *inspectable*.

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